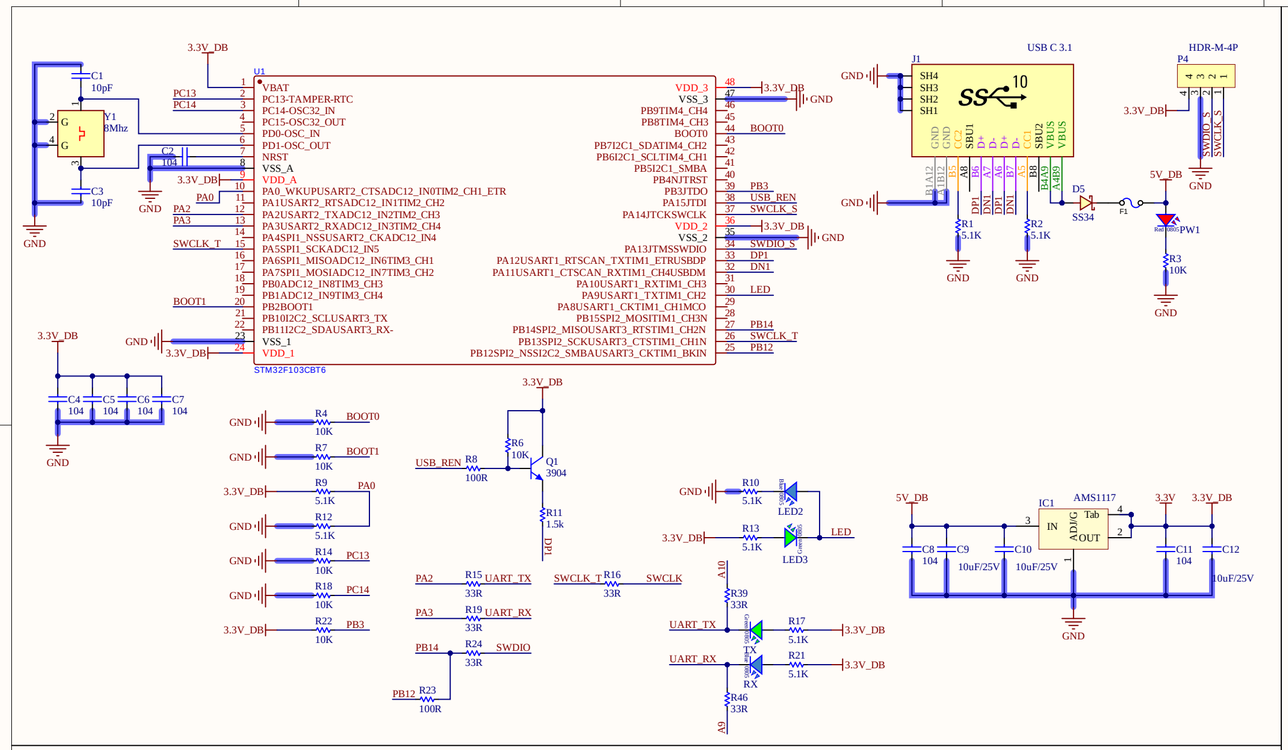
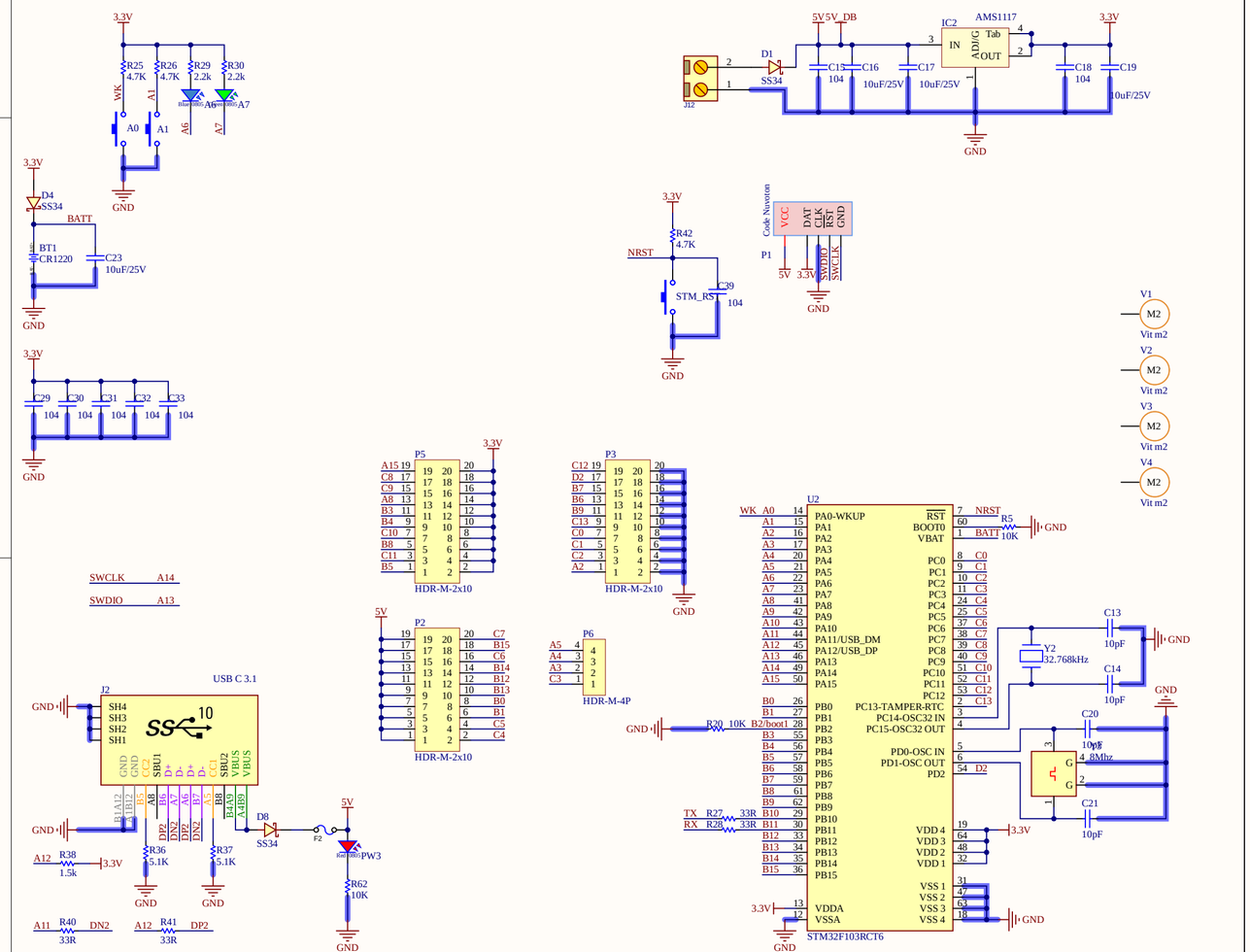
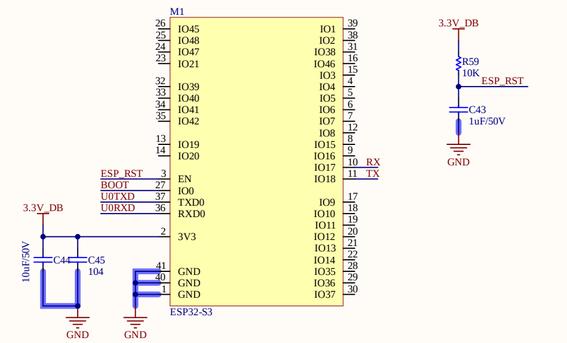

3.3V_DB
U1
STM32F103CBT6
VBAT
PC13-TAMPER-RTC
PC14-OSC32_IN
PC15-OSC32_OUT
PD0-OSC_IN
PD1-OSC_OUT
NRST
VSS_A
VDD_A
PA0_WKUPUSART2_CTSADC12_IN0TIM2_CH1_ETR
PA1USART2_RTSADC12_IN1TIM2_CH2
PA2USART2_TXADC12_IN2TIM2_CH3
PA3USART2_RXADC12_IN3TIM2_CH4
PA4SPI1_NSSUSART2_CKADC12_IN4
PA5SPI1_SCKADC12_IN5
PA6SPI1_MISOADC12_IN6TIM3_CH1
PA7SPI1_MOSIADC12_IN7TIM3_CH2
PB0ADC12_IN8TIM3_CH3
PB1ADC12_IN9TIM3_CH4
PB2BOOT1
PB10I2C2_SCLUSART3_TX
PB11I2C2_SDAUSART3_RX-
VSS_1
VDD_1
VDD_3
VSS_3
PB9TIM4_CH4
PB8TIM4_CH3
BOOT0
PB7I2C1_SDATIM4_CH2
PB6I2C1_SCLTIM4_CH1
PB5I2C1_SMBA
PB4NJTRST
PB3JTDO
PA15JTDI
PA14JTCKSWCLK
VDD_2
VSS_2
PA13JTMSSWDIO
PA12USART1_RTSCAN_TXTIM1_ETRUSBDP
PA11USART1_CTSCAN_RXTIM1_CH4USBDM
PA10USART1_RXTIM1_CH3
PA9USART1_TXTIM1_CH2
PA8USART1_CKTIM1_CH1MCO
PB15SPI2_MOSITIM1_CH3N
PB14SPI2_MISOUSART3_RTSTIM1_CH2N
PB13SPI2_SCKUSART3_CTSTIM1_CH1N
PB12SPI2_NSSI2C2_SMBAUSART3_CKTIM1_BKIN
C1 10pF
C2 104
C3 10pF
Y1 8Mhz
C4 104
C5 104
C6 104
C7 104
R4 10K
R7 10K
R9 5.1K
R12 5.1K
R14 10K
R18 10K
R22 10K
R6 10K
Q1 3904
R8 100R
R11 1.5k
R10 5.1K
R13 5.1K
LED2
LED3
R15 33R
R19 33R
R24 33R
R23 100R
R16 33R
R39 33R
R17 5.1K
R21 5.1K
R46 33R
J1
USB C 3.1
R1 5.1K
R2 5.1K
D5
SS34
F1
PW1
R3 10K
P4
HDR-M-4P
SWDIO_S
SWCLK_S
5V_DB
GND
IC1
AMS1117
C8 104
C9 10uF/25V
C10 10uF/25V
C11 104
C12 10uF/25V
BOOT0
BOOT1
PA0
PA2
PA3
PB3
PB14
PB12
PC13
PC14
SWCLK_T
USB_REN
UART_TX
UART_RX
SWDIO
SWCLK
LED
DP1
DN1
DP2
DN2


3.3V
R25 4.7K
R26 4.7K
R29 2.2k
R30 2.2k
WK
A1
A6
A7
A0
A1
GND
D4
SS34
BATT
BT1
CR1220
C23 10uF/25V
C29 104
C30 104
C31 104
C32 104
C33 104
R42 4.7K
NRST
STM_RST
C39 104
P1
Code Nuvoton
VCC
DAT
CLK
RST
GND
SWDIO
SWCLK
5V
3.3V
V1
V2
V3
V4
M2
Vit m2
P5
HDR-M-2x10
P3
HDR-M-2x10
P2
HDR-M-2x10
P6
HDR-M-4P
SWCLK A14
SWDIO A13
5V
USB C 3.1
J2
D8
SS34
F2
PW3
R62 10K
R38 1.5k
R36 5.1K
R37 5.1K
R40 33R
R41 33R
U2
STM32F103RCT6
WK A0
PA0-WKUP
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11/USB_DM
PA12/USB_DP
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13-TAMPER-RTC
PC14-OSC32 IN
PC15-OSC32 OUT
PD0-OSC IN
PD1-OSC OUT
PD2
RST
BOOT0
VBAT
VDD 4
VDD 3
VDD 2
VDD 1
VSS 1
VSS 2
VSS 3
VSS 4
VDDA
VSSA
R5 10K
R20 10K
B2/boot1
TX R27 33R
RX R28 33R
Y2 32.768kHz
C13 10pF
C14 10pF
C20 10pF
C21 10pF
8Mhz
IC2
AMS1117
C13 104
C16 10uF/25V
C17 10uF/25V
C18 104
C19 10uF/25V
J12
D1
SS34

J1
USB C 3.1
SH4
SH3
SH2
SH1
GND
GND
CC2
SBU1
D+
D+
D-
D-
SBU2
VBUS
VBUS
B1A12
B12
B5
A8
B6
A7
A6
B7
A5
B8
B4A9
A4B9
R1 5.1K
R2 5.1K
D5
SS34
F1
PW1
R3 10K
P4
HDR-M-4P
SWDIO_S
SWCLK_S
5V_DB
GND

5V_DB
IC1
AMS1117
IN
ADJ/G
Tab
OUT
3.3V
3.3V_DB
C8 104
C9 10uF/25V
C10 10uF/25V
C11 104
C12 10uF/25V
GND

5V_DB
R47 10K
PW2
GND
U5
CH340_D+
CH340_D-
D+
D-
RXD
TXD
RTS#
CTS#
TNOW/DTR#
VCC
GND
V3
CH340X
CH340_RX
CH340_TX
ESP_RST
BOOT
1N4148
R49 33R
C40 104
C41 104
3.3V_DB
J9
USB C 3.1
VBUS
SBU2
CC1
D-
D+
SBU1
CC2
A4B9
B4A9
B8
A5
B7
A6
A7
B6
A8
B5
A1B12
B1A12
R48 5.1K
R50 5.1K
D6
SS34
CH340_D-
CH340_D+
CH340_RX
R52 499R
U0TXD
CH340_TX
R56 499R
U0RXD
BOOT
ESP_RST


M1
ESP32-S3
IO45
IO48
IO47
IO21
IO39
IO40
IO41
IO42
IO19
IO20
IO1
IO2
IO38
IO46
IO3
IO4
IO5
IO6
IO7
IO8
IO15
IO16
IO17
IO18
IO9
IO10
IO11
IO12
IO13
IO14
IO35
IO36
IO37
EN
IO0
TXD0
RXD0
3V3
GND
ESP_RST
BOOT
U0TXD
U0RXD
RX
TX
3.3V_DB
R59 10K
C43 1uF/50V
C44
C45 104
10uF/50V